NYC Motor Vehicle Collision Analysis

Group 3-
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Presentation Outline

01 **Dataset Overview**
NYPD schema, coverage, and data quality audit

02 **Data Pipeline**
7-stage ingestion and validation architecture

03 **Multi-Year Trends**
Collision volume and fatality rate over time

04 **Temporal Patterns**
Hourly, weekday, and borough breakdown

05 **Contributing Factors**
Top causes, vehicle types, weekend vs weekday

06 **COVID-19 Impact**
2020 phase-by-phase analysis

07 **Severity & Risk**
Rush hours, lethality matrix, DST effects

08 **Spatial Clustering**
NYC hotspot detection — 140 clusters

09 **Predictive Modeling**
Logistic regression injury/fatality baseline

10 **Conclusions**
Key findings and takeaways

Dataset Overview — NYPD Motor Vehicle Collisions

2,246,476
Total Collision Records
Full multi-year dataset

13+ Years
Date Coverage
Jul 2012 – Mar 2026

29
Schema Columns
Per record

74,881
2020 COVID Snapshot
Jan – Aug 2020

About the Data

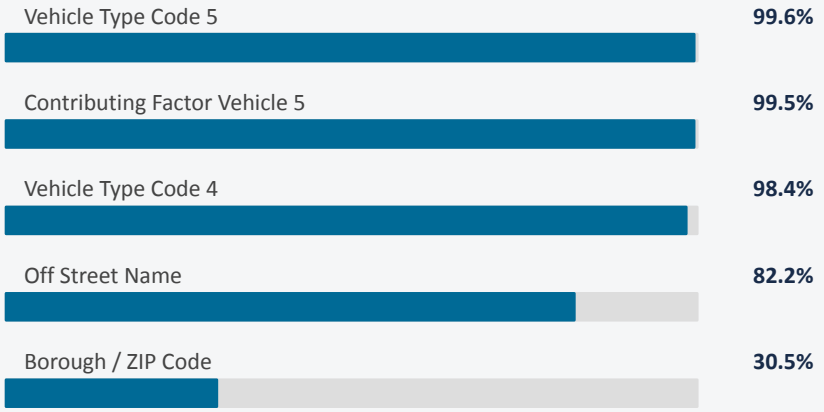
Official Source: NYPD Motor Vehicle Collision records from NYC Open Data portal

29 Columns: Includes date/time, borough, ZIP code, GPS coordinates, injury & fatality counts by person type, contributing factors (up to 5 per crash), and vehicle types (up to 5 per crash)

Data Quality: Zero invalid crash-date rows and zero duplicate collision IDs — confirmed by automated QA stage

Two Configs: Pipeline runs in two modes: full multi-year trend analysis and 2020 COVID phase snapshot

Top Missing Fields

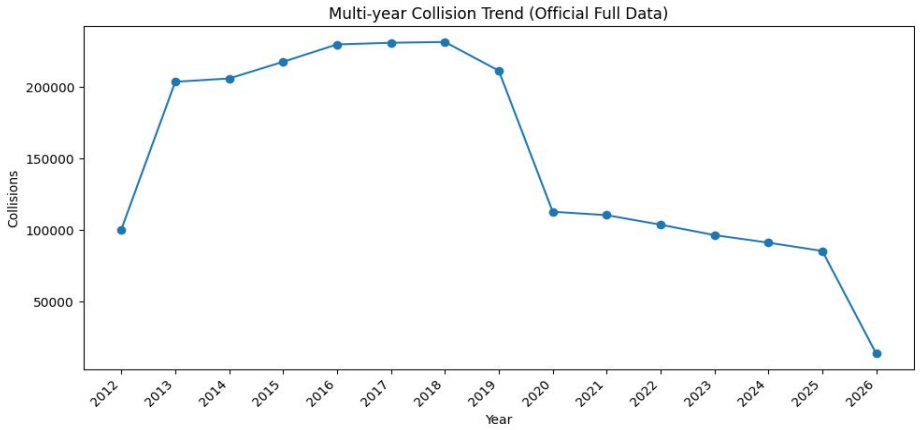


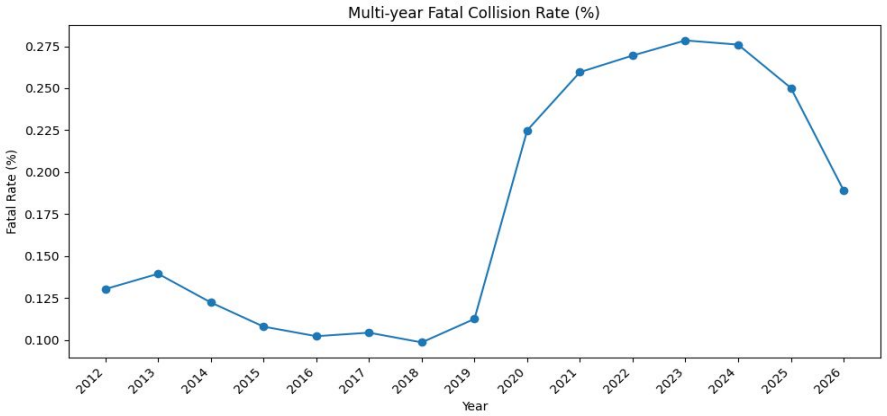
Data Pipeline — 7-Stage Sequential Architecture

1. **Data Ingestion**
   Load CSV, normalize column aliases, parse dates → 2,246,476 canonical rows
2. **Data Quality**
   Validate schema, detect duplicates, profile missing values, generate QA report
3. **Feature Engineering**
   Derive hour, day_of_week, is_weekend, time_block, pandemic_era, dst_transition
4. **Exploratory Analysis**
   KPIs by year/month/hour/borough; trend charts, factor & vehicle distributions
5. **Severity Modeling**
   Logistic regression (from stdlib) — predict any_injury & fatal_collision; ROC/PR curves
6. **Spatial Clustering**
   MiniBatchKMeans on GPS coords (from stdlib) — 140 clusters; hotspot maps
7. **Validation Tests**
   Cross-check all output totals vs canonical dataset — 15/15 checks passed

✓ All 15 validation checks passed · Only external dependency: matplotlib · All ML & clustering from Python standard library

Multi-Year Collision Trends (2012–2026)



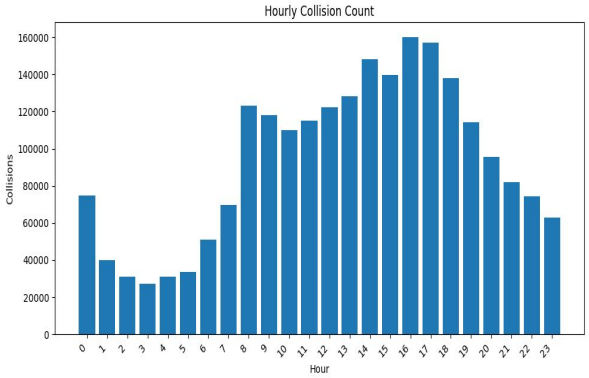


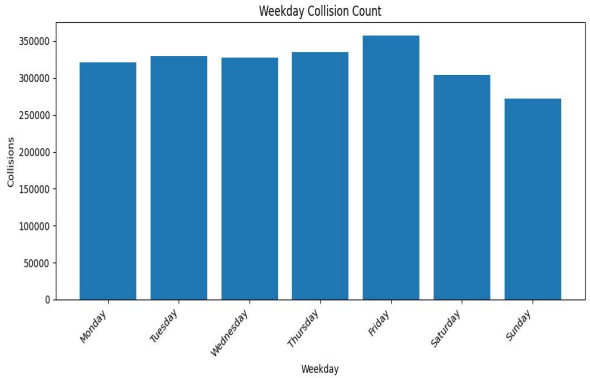
Peak volume: 2017–2018, followed by steady decline through 2022

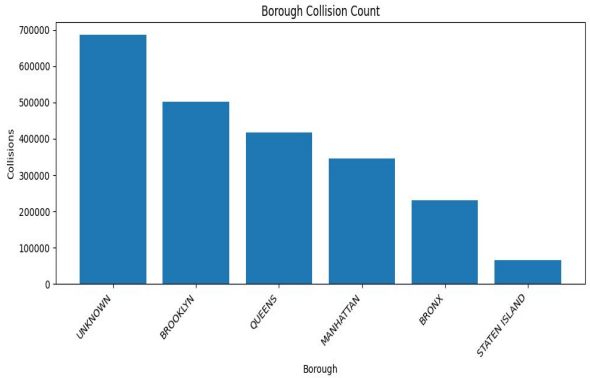
COVID 2020: Sharpest single-year volume drop — down ~45% from prior year

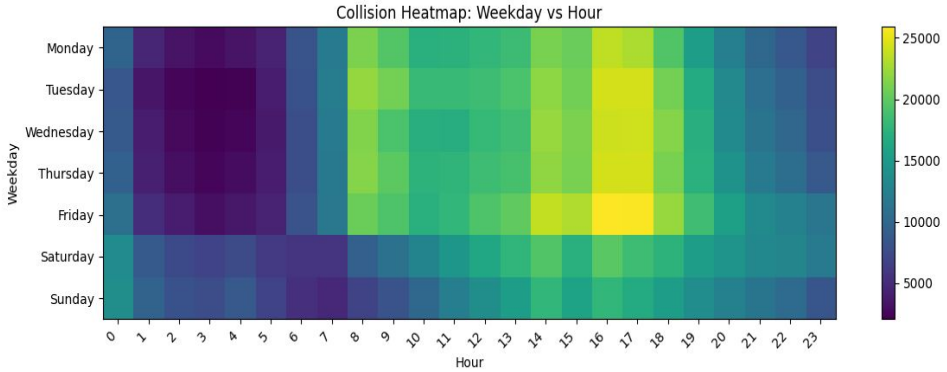
Fatal rate: Rose during pandemic — fewer cars, higher speeds on empty streets

Temporal & Geographic Patterns









Key Insights

Rush hours: 8–9 AM and 4–6 PM peak

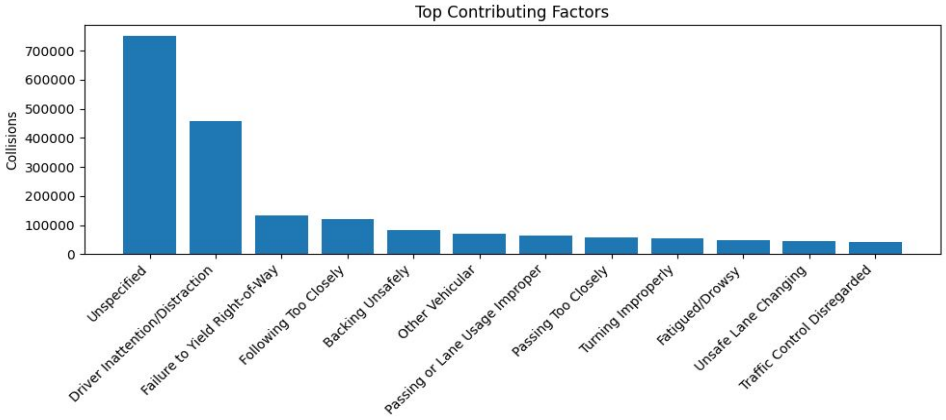
Deadliest: 1–5 AM — low volume, high risk

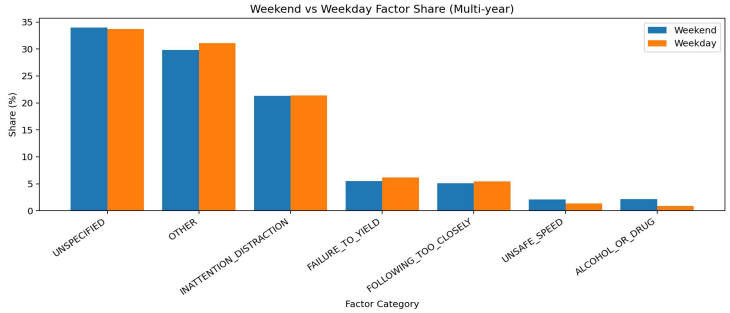
Busiest day: Friday leads all weekdays

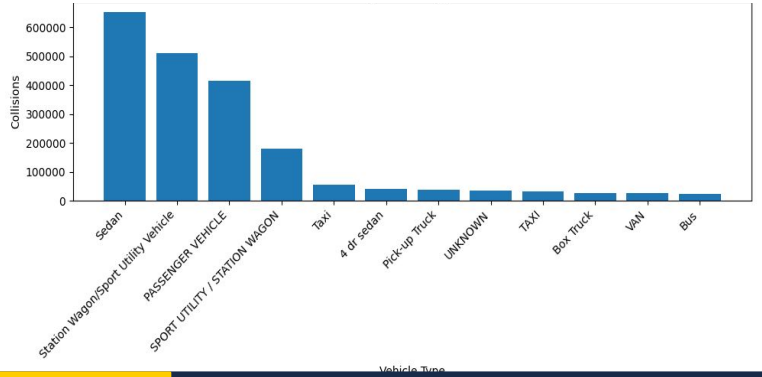
Top borough: Brooklyn has most collisions

Heatmap: Friday evenings densest risk zone

Contributing Factors & Vehicle Types





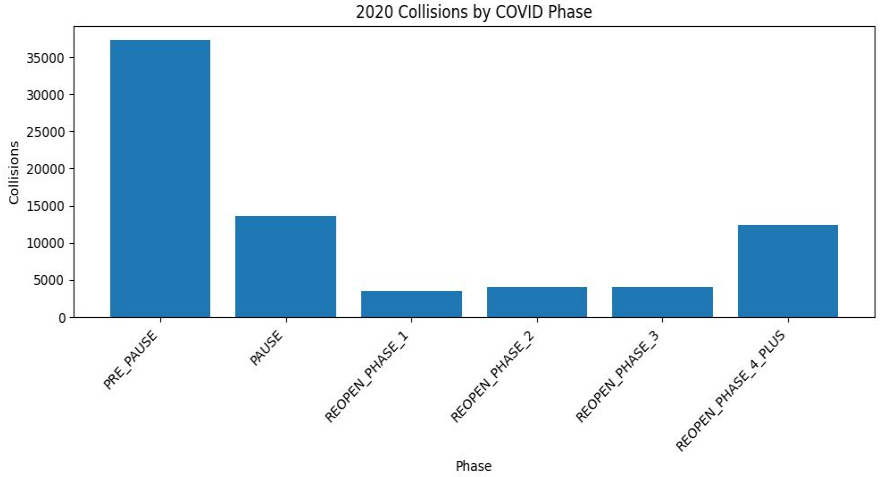


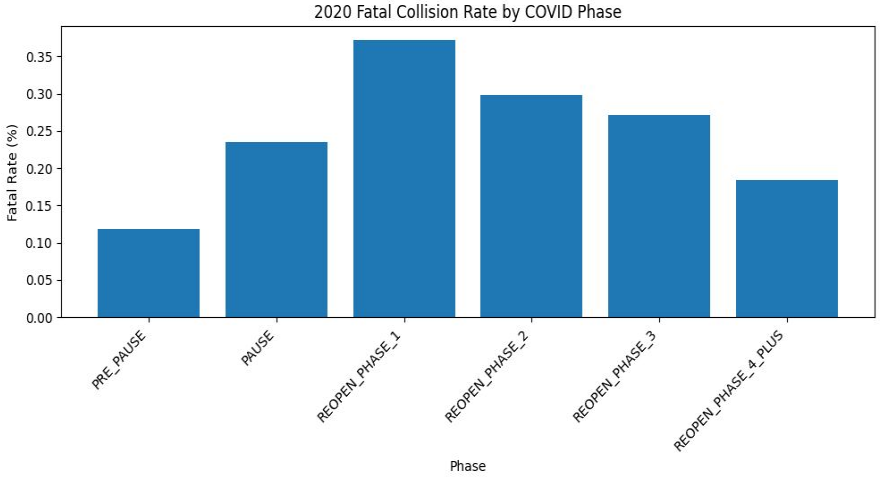
Top Factors

- #1 Driver Inattention / Distraction
- #2 Failure to Yield Right-of-Way
- #3 Following Too Closely
- Unsafe Speed spiked post-2020
- Passenger vehicles dominate all types

COVID-19 Impact — 2020 Phase Segmentation

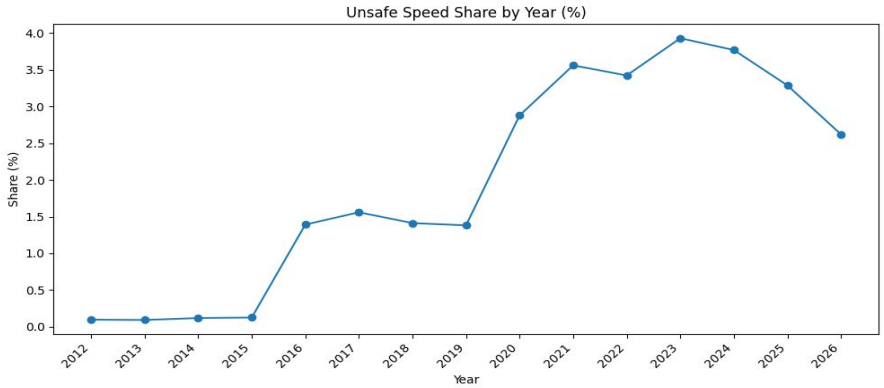
Pre-PAUSE	PAUSE	Reopen Ph.1	Reopen 2–4
Jan 1 – Mar 21	Mar 22 – Jun 7	Jun 8 – Jun 21	Jun 22 – Aug 29
37,269	**13,590**	**3,496**	**20,526**
collisions	collisions	collisions	collisions
Fatal: 0.12% | Killed/10k: 12.6	Fatal: 0.24% | Killed/10k: 24.3	Fatal: 0.37% | Killed/10k: 42.9	Fatal: 0.22% | Killed/10k: 22.1

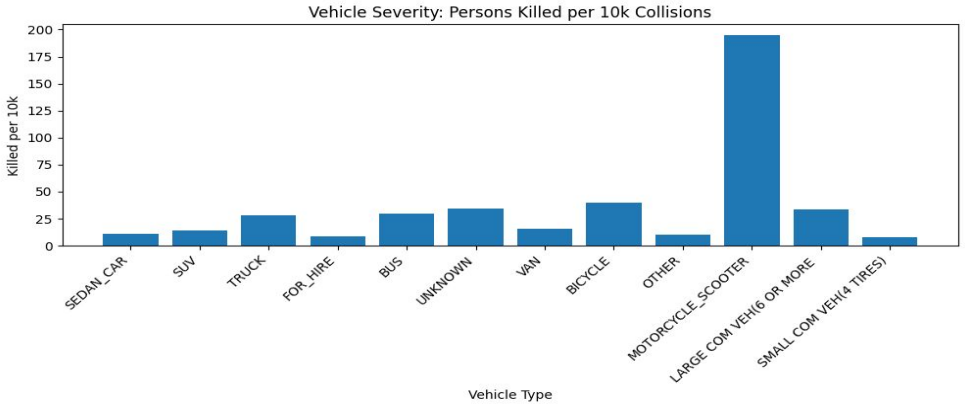


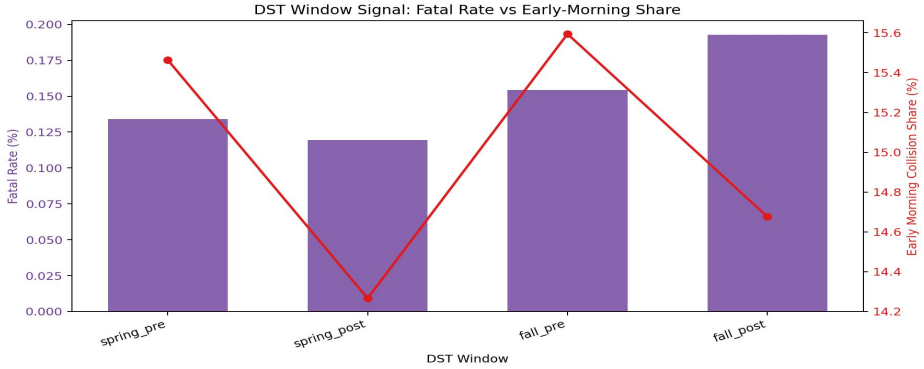


PAUSE cut collisions 64% vs Pre-PAUSE — but fatal rate doubled (0.12% → 0.24%), peaking at 0.37% in Reopen Phase 1

Severity & Risk — Rush Hours, Vehicles & DST Effects



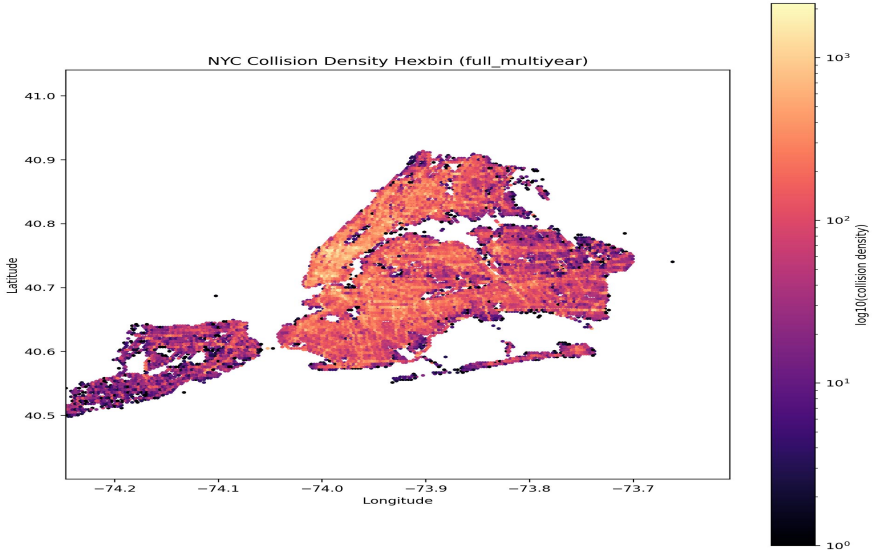


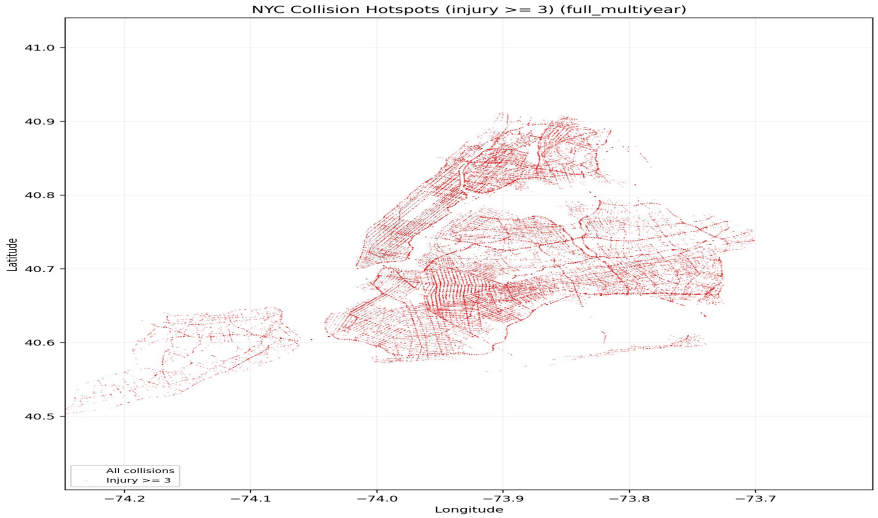


Key Insights

- Unsafe speeds skyrocketed post COVID
- Unsafe speeds tapered down few years post COVID
- Motorcycles/Scooters most at risk of fatality on roads
- Post Daylight Savings Time shift in fall, there is a surge in overall fatalities due to accidents at all times of the day
- Morning fatalities are generally lower post DST in both spring and fall

Spatial Clustering — NYC Collision Hotspots



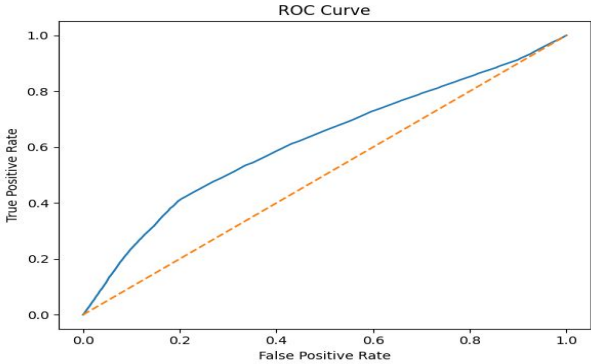


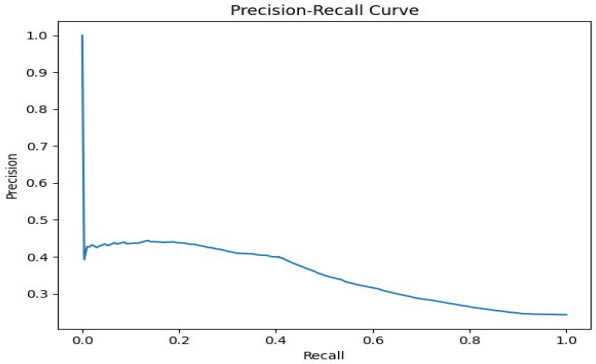
2,005,851	140	MiniBatch KMeans
GPS-valid records	Spatial clusters	Algorithm (stdlib)

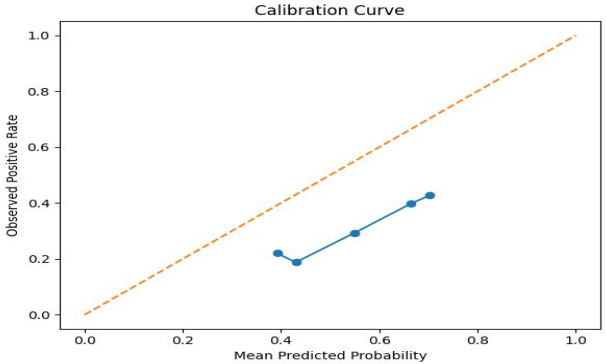
Density hexbins confirm Manhattan midtown and Brooklyn corridors as highest density zones
Road imprint scatter (right) reveals NYC's street grid embedded in collision data - validating GPS quality

Predictive Modeling — Logistic Regression Baseline

Metric	Full Multi-Year (250k train rows)	2020 Snapshot (59.8k train rows)
Accuracy	**70.45%**	59.16%
ROC-AUC	**0.619**	0.591
PR-AUC	**0.350**	0.341
F1 Score	**0.405**	0.414







Conclusions & Key Findings

2.24M records analyzed — clean, reproducible pipeline

Zero duplicates or invalid dates. All 15 validation checks passed across both run configurations.

COVID-19 caused the sharpest traffic shift ever recorded in the dataset

PAUSE reduced volume 64% but doubled the fatal rate, which peaked at 0.37% in Reopen Phase 1.

Daylight savings time shift increases fatalities in fall

Fatalities are lower post DST in Spring, but significantly higher post DST in fall – changes in sleep/schedule affects drivers

Spatial clustering reveals persistent NYC hotspots

140 clusters on 2M GPS-valid records. Brooklyn corridors and Manhattan midtown are consistently most dense.

Logistic model achieves ROC-AUC 0.619 on full dataset

Reasonable baseline from Python stdlib only. Richer features (road type, weather) could drive improvement.